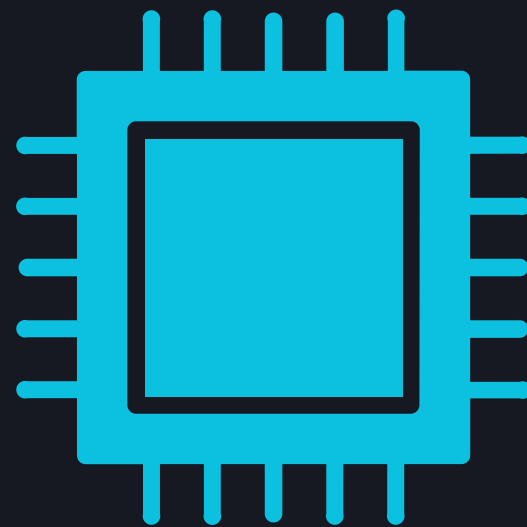

Efficient VLSI Design-Based Adaptive Pipelined RISC-V Core *for Low-Power Edge Devices*

LIEW MING HENG

Bachelor of Electronic Engineering | Universiti Sains Malaysia | 2026

Supervisor: Ts. Dr. Siti Fatimah Abd Rahman





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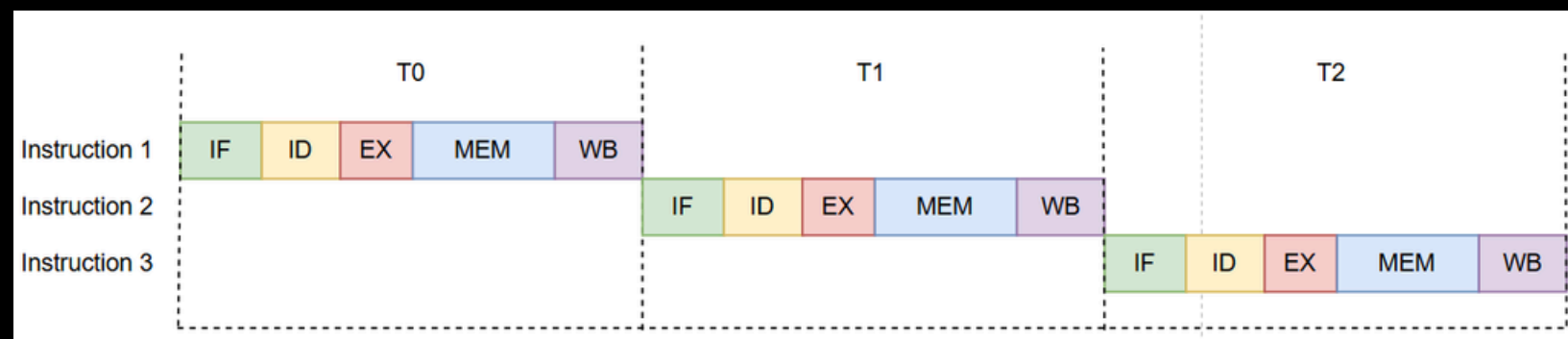
CONCLUSION

RESEARCH BACKGROUND



RESEARCH BACKGROUND

SINGLE-CYCLE PROCESSOR

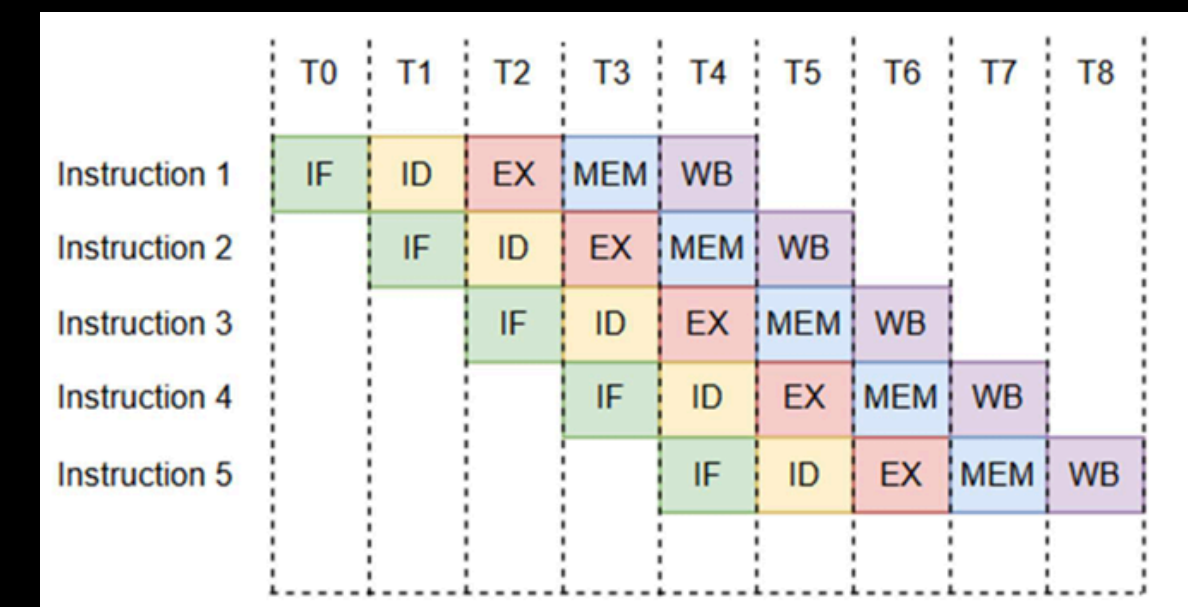


IF : INSTRUCTION FETCH
ID : INSTRUCTION DECODE
EX : EXECUTE
MEM : MEMORY
WB : WRITE-BACK

PIPELINING TECHNIQUES

- *Increases maximum operating frequency*

5-STAGE PROCESSOR



RESEARCH BACKGROUND

DYNAMIC POWER EQUATION

$$P_{dyn} = \alpha \cdot C \cdot V^2 \cdot f$$

- α : Switching Activity Factor
- C : Effective Load Capacitance
- V : Supply Voltage
- f : Operating Frequency

PROBLEM STATEMENT





PROBLEM STATEMENT

PROBLEM 1: RIGID PIPELINE TRADE-OFF & HAZARD PENALTY



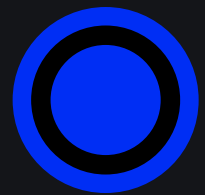
SHALLOW PIPELINE

- Lower clock frequency due to a longer critical path per stage.
- Lower throughput that limits performance for complex computational workloads.



DEEP PIPELINE

- Higher clock frequency and peak performance enabled by shorter stage propagation delays.
- Triggers costly pipeline flushes and stalls
- Higher dynamic power consumption



PROBLEM STATEMENT

PROBLEM 2: WORKLOAD-ARCHITECTURE MISMATCH

FIXED 5-STAGE PIPELINE



High-Complexity Instructions



Optimal Power & Peak Performance



Low-Complexity Instructions



Power Inefficiency

Wasted Dynamic Power
in Unused Stages



Redundant Clock Switching



RESEARCH OBJECTIVES



Research Objectives

1

Design a *runtime-reconfigurable 3/5-stage RV32I processor* that saving $\geq 50\%$ dynamic power in 3-stage mode.

2

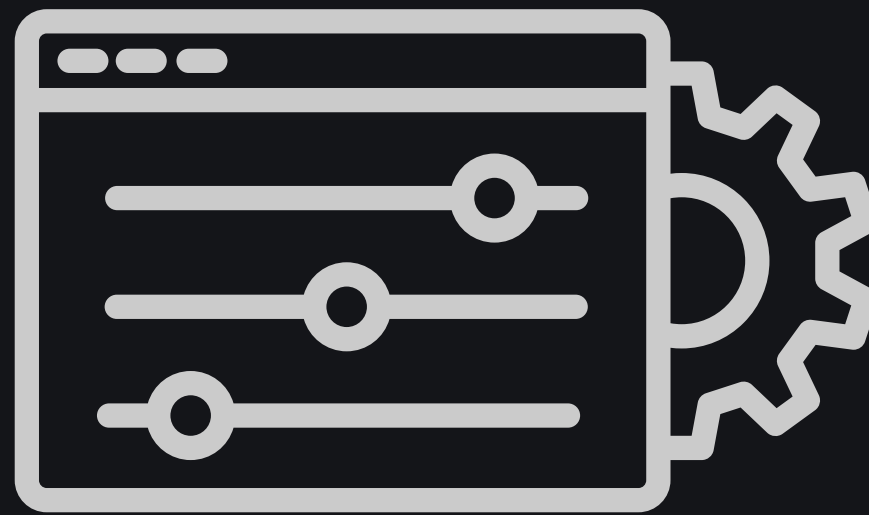
Achieve *100% ISA compliance* across all 39 RISCOF architectural tests.

3

Deploy the design on a Lattice CPNX-Versa FPGA, ensuring target clock frequencies of *50MHz (5-stage mode)* and *20MHz (adaptive 3-stage mode)*.

Existed Research vs Proposed Work

	Minimalist Soft-Cores PicoRV32	High-Performance Rocket SSR	Proposed Work
Pipeline Depth	Rigid Multi-Cycle	Rigid Fixed 5-Stage	Runtime-Reconfigurable
Hazard Strategy Overhead	Zero	High Power Penalty	Dynamic Elimination via Stage Unification
Verification	Local Testbenches	RISCOF	RISCOF
Application	FPGA Soft-Core	ASIC	Lattice CPNX FPGA



METHODOLOGY

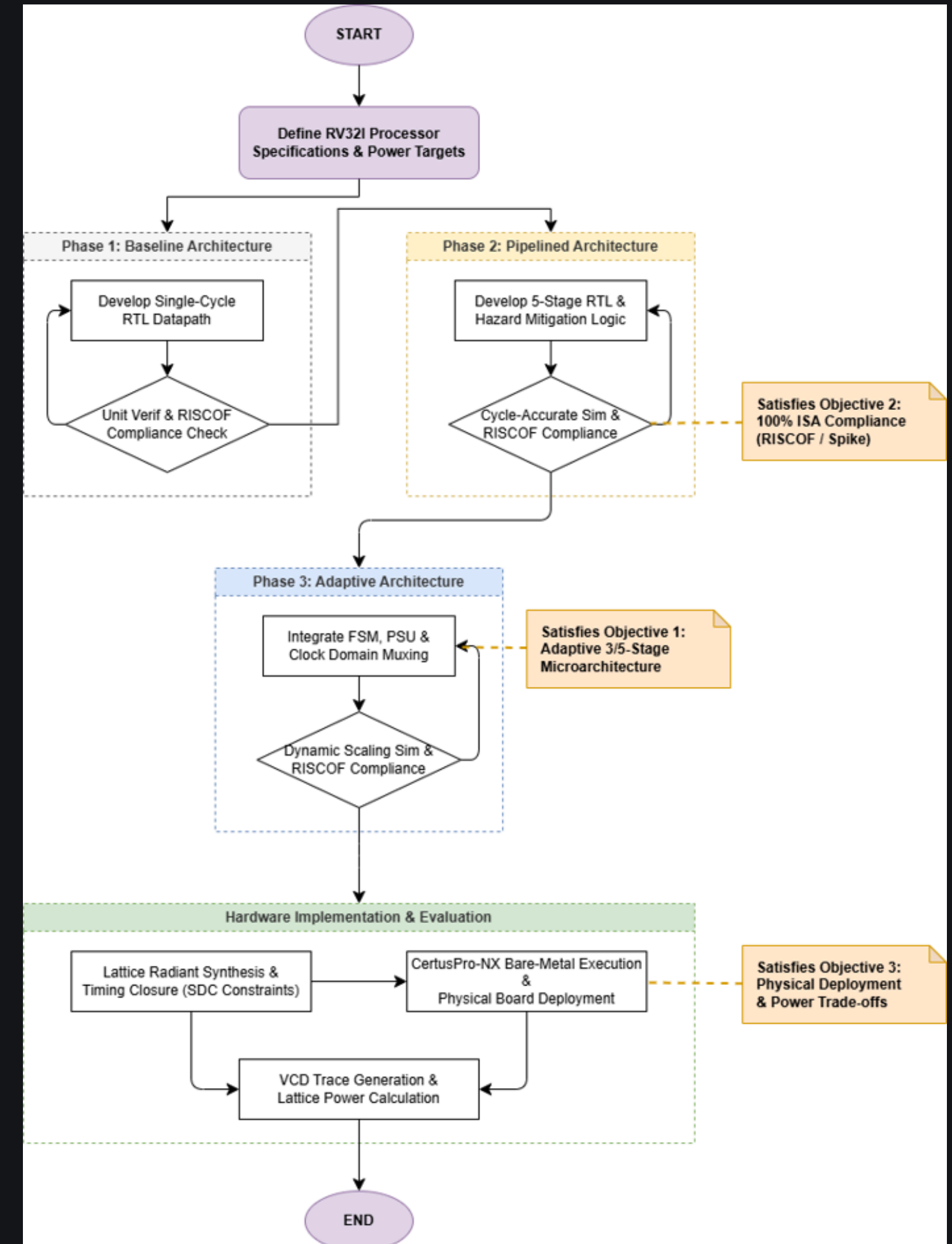
METHODOLOGY

The process is segmented into 3 primary phases that govern the lifecycle of the core development.

PHASE 1 BASELINE ARCHITECTURE

PHASE 2 PIPELINED ARCHITECTURE

PHASE 3 ADAPTIVE ARCHITECTURE

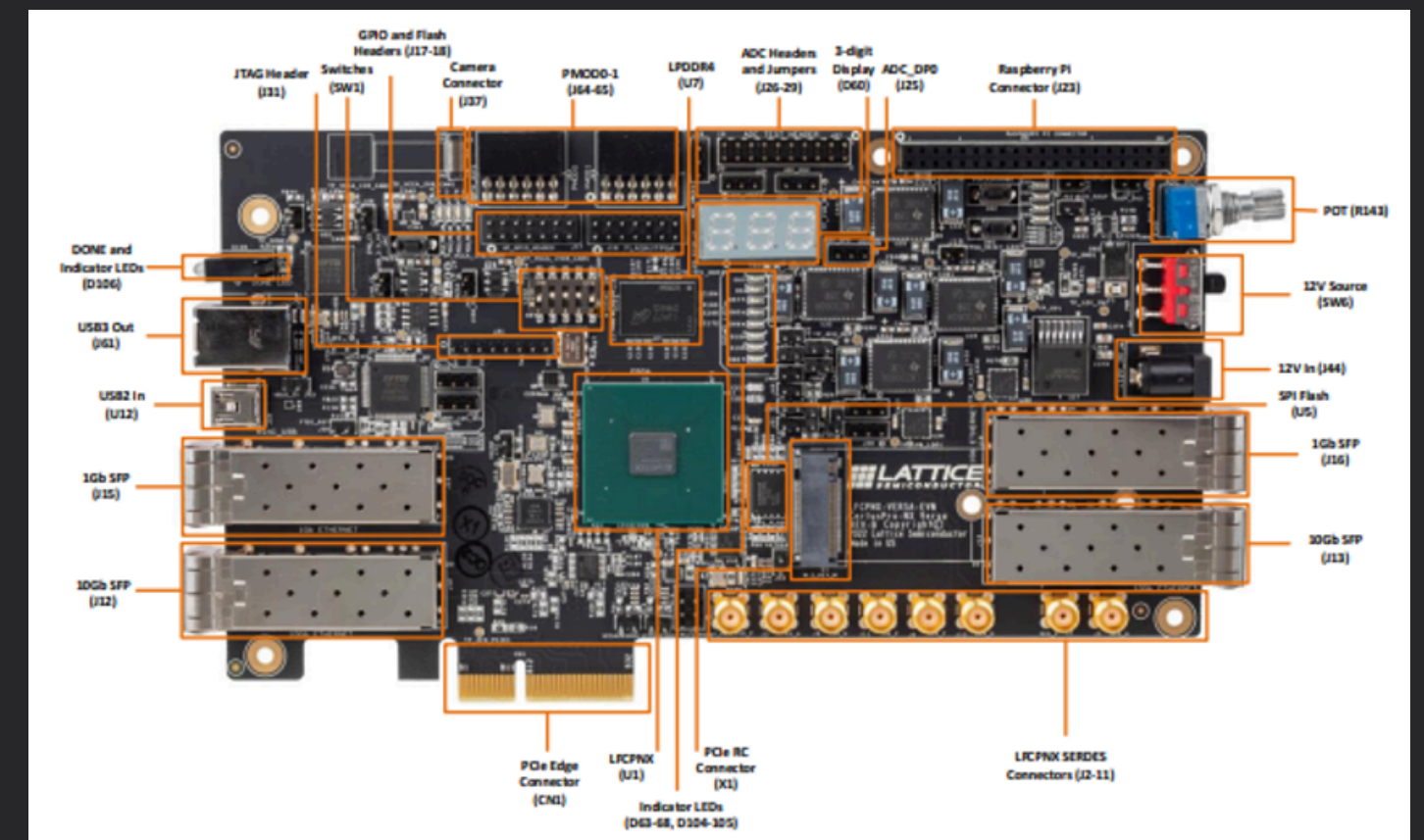


Project Resources

SOFTWARE

Category	Tool	Purpose
HDL	Verilog-2001	RTL modeling of the 3-to-5 stage core.
Simulation	Questasim / Verilator	Functional verification and .vcd files generation.
Synthesis	Lattice Radiant (LSE)	Logic mapping and timing closure for CPNX-Versa FPGA board.
Compliance	RISCOF + Spike	“Golden Model” verification for ISA compliance.
Toolchain	RISC-V GNU Toolchain	Compiling C codes to machine codes.

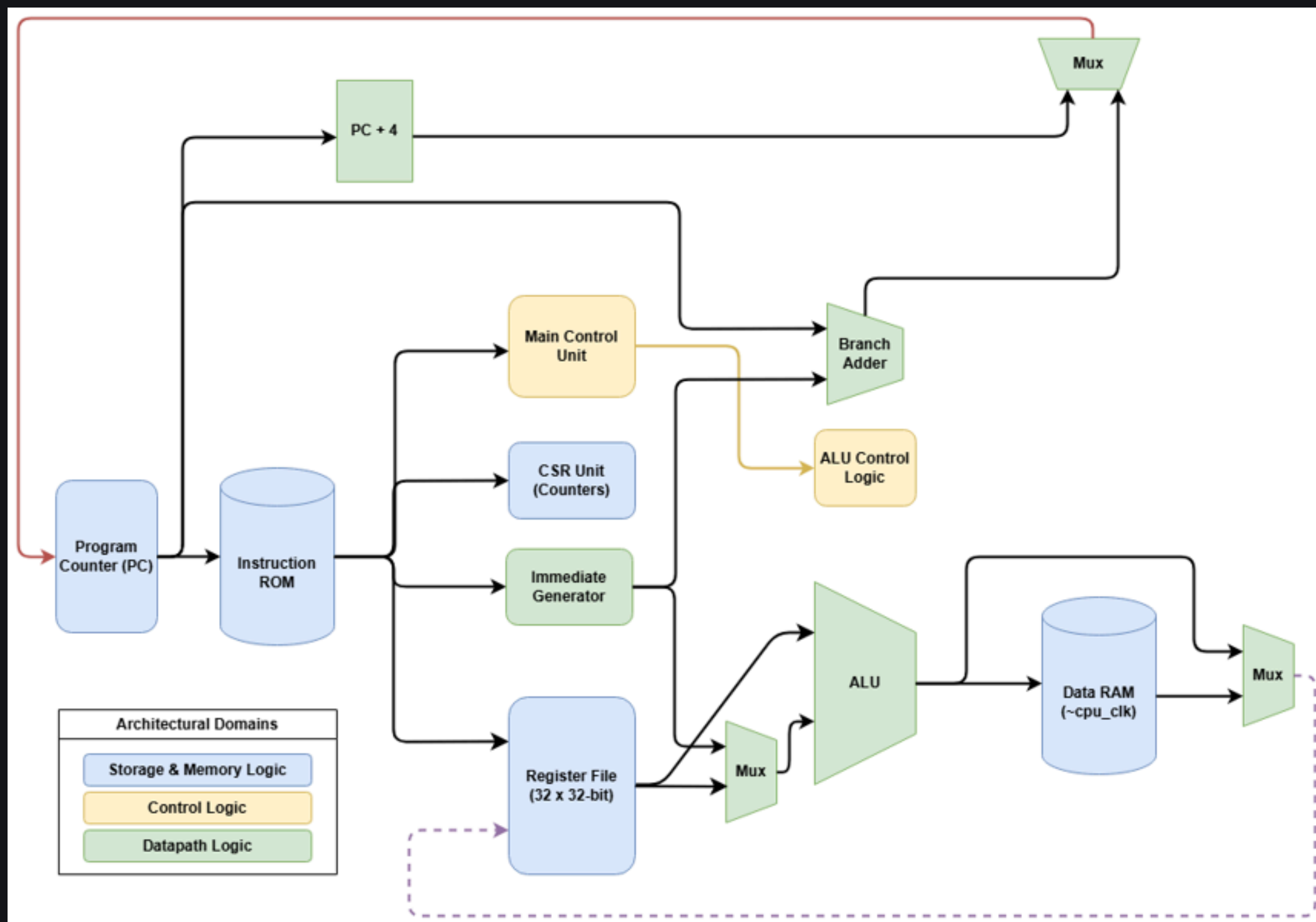
HARDWARE



Lattice CPNX Versa FPGA Board



PHASE 1



Microarchitectural of the
Baseline Single-Cycle RV32I Core

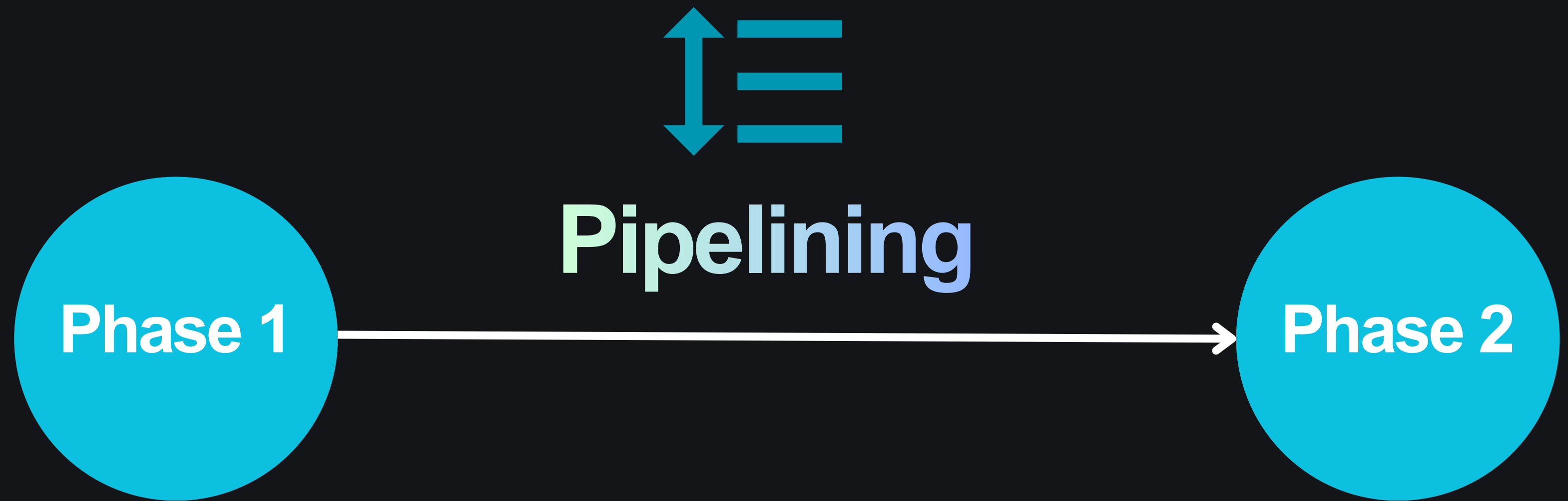
Own Design

- Control Unit
- ALU Control
- Arithmetic Logic Unit (ALU)
- Immediate Generator
- Program Counter (PC)
- Register File
- Control and Status Register (CSR) Unit
- Instruction ROM

Lattice IP

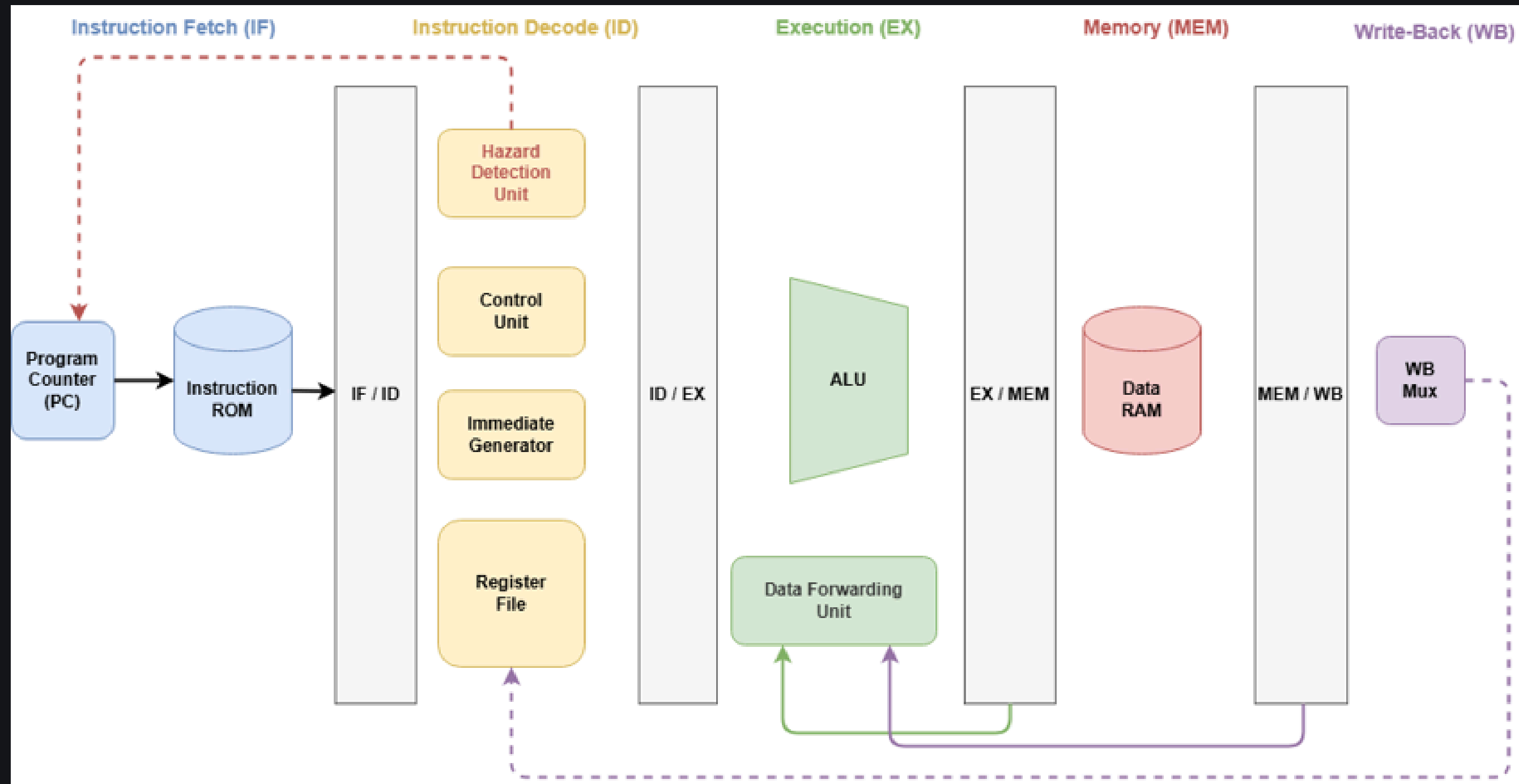
- Data RAM

Higher resource efficiency (minimizing the use of LUT4 logic)





PHASE 2



Microarchitectural of the
5-Stage Pipelined RV32I Core



PHASE 2

Hazard Mitigation Logics

1

Data Forwarding Unit

2

Hazard Detection Unit

3

Pipeline Flushing

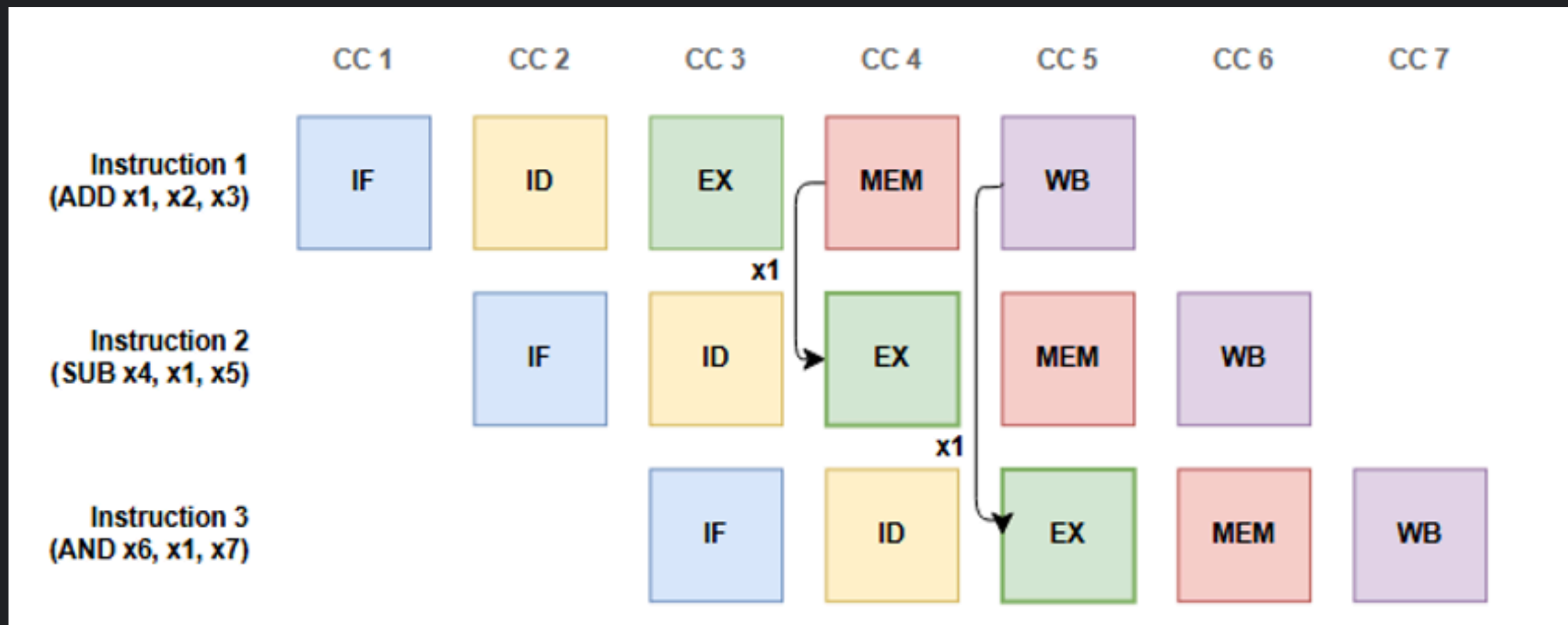


PHASE 2

Hazard Mitigation Logics

1

Data Forwarding Unit



- Resolving Read-After-Write (RAW) Hazards

RAW Hazards

An instruction in the ID stage requires an operand that is still in progress.

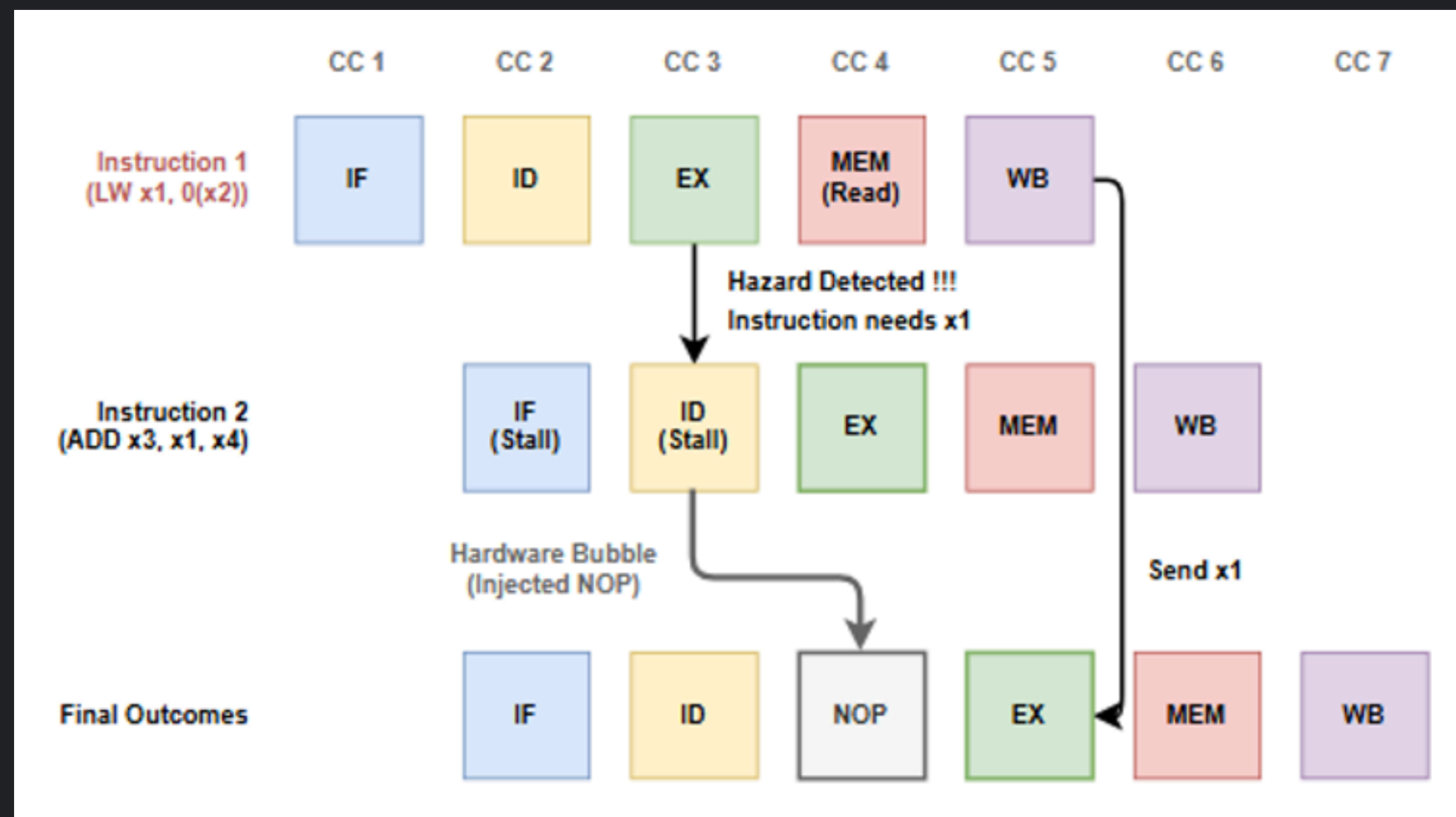


PHASE 2

Hazard Mitigation Logics

2

Hazard Detection Unit



- Resolving Load-Use Stalls

Load-Use Stalls

An instruction attempts to use a register immediately after a memory load targets that same register.

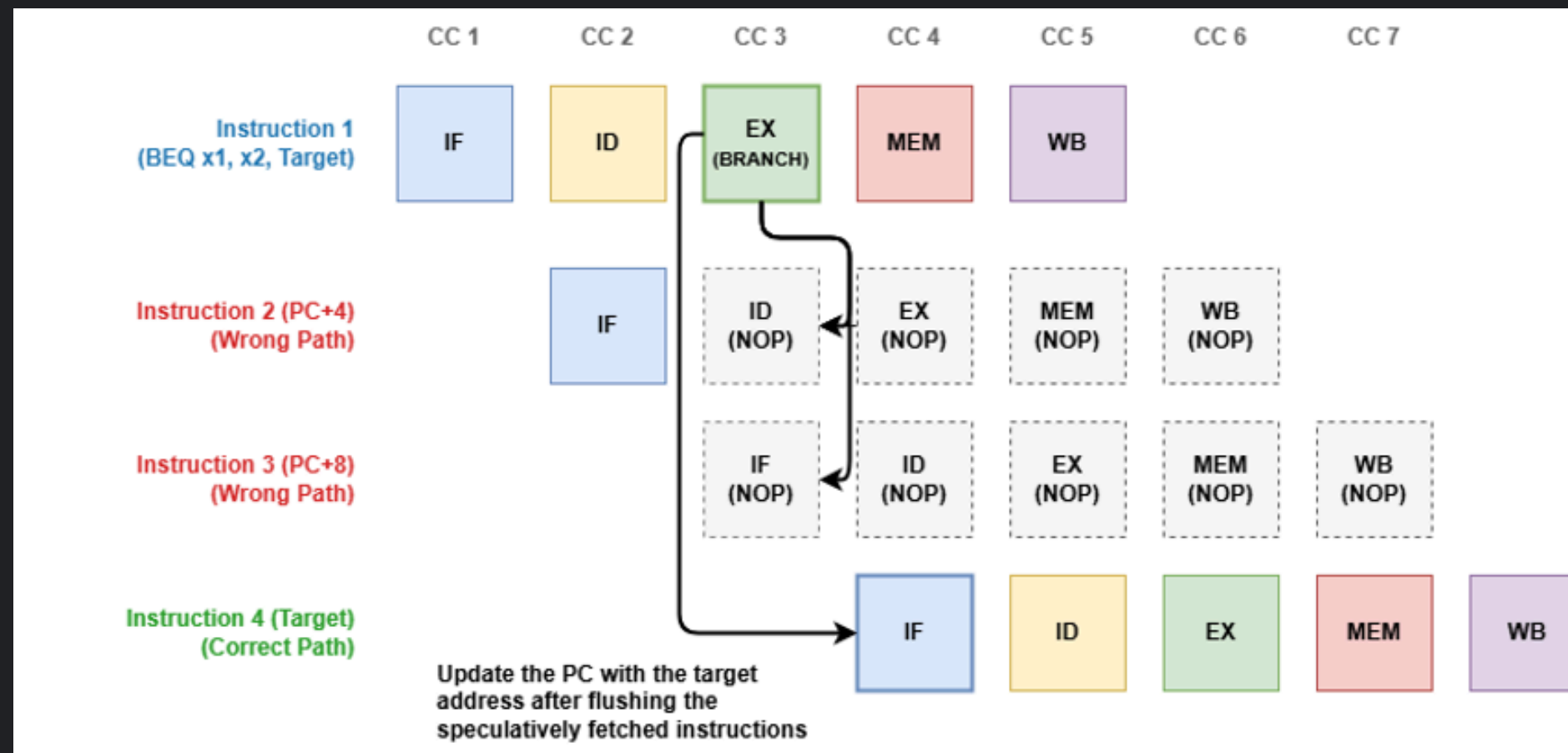


PHASE 2

Hazard Mitigation Logics

3

Pipeline Flushing



- Resolving Branch Control Hazard

Branch Control Hazard

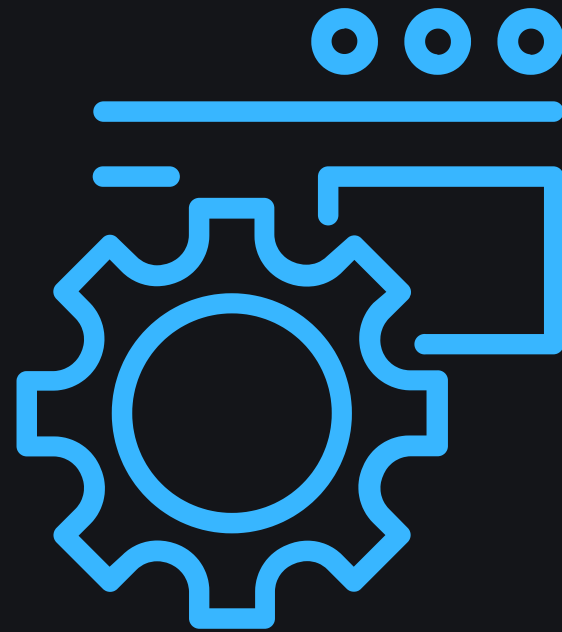
Control hazards happen when a branch instruction forces the processor to guess and fetch the next two sequential instructions before knowing if the branch is actually taken.

If the branch evaluates as “taken” during the Execution (EX) stage, those speculatively fetched instructions are invalid and must be discarded from the pipeline.

Software- Reconfigurable

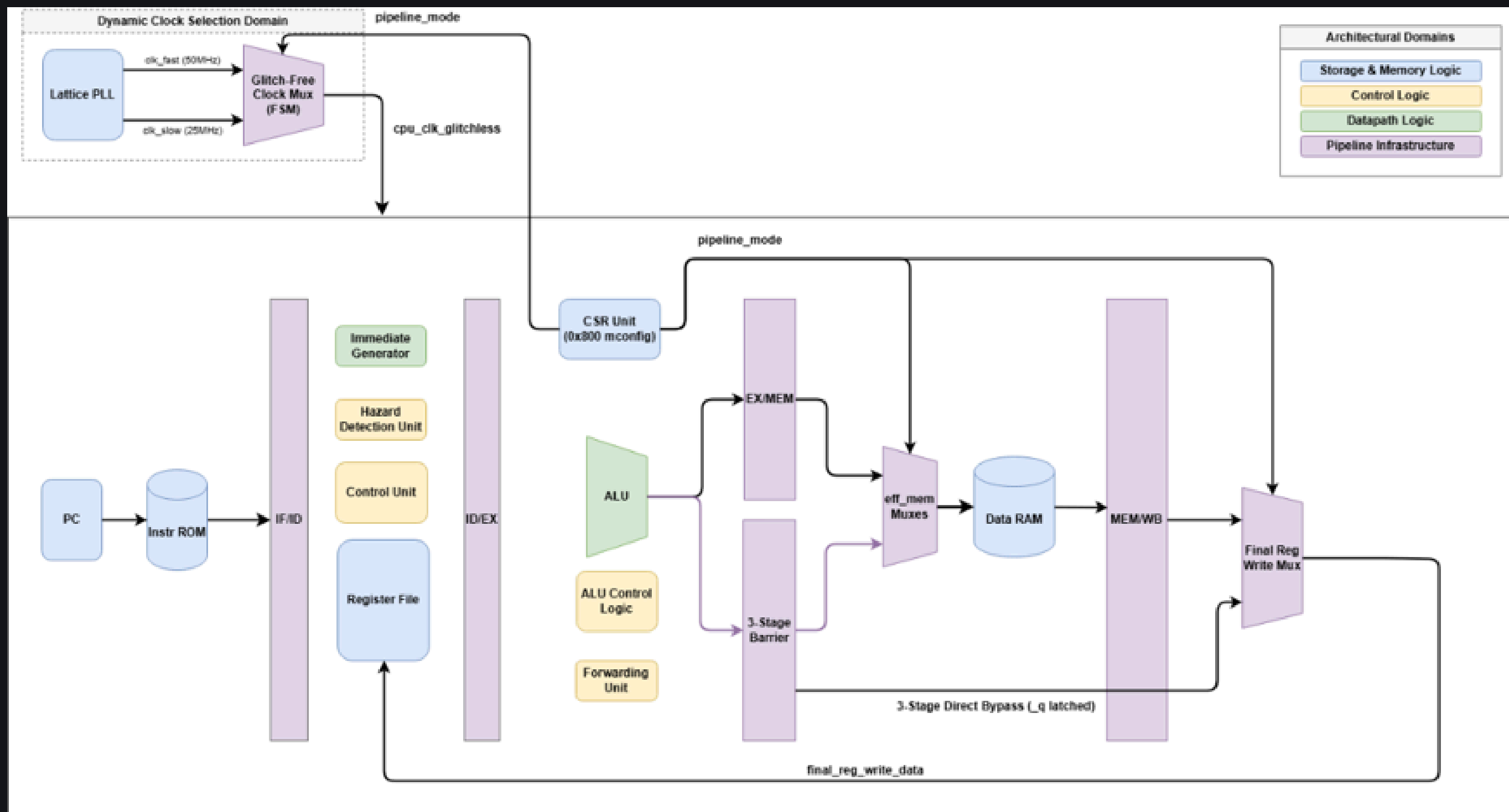
Phase 2

Phase 3





PHASE 3



Microarchitectural of the
Adaptive RV32I Core

Dual-Clock Domain Architecture

A dual-domain clocking strategy is required because the 5-stage mode demands 50 MHz for high performance, while the 3-stage mode must run at 20 MHz to ***avoid timing violations and maximize power savings by 60%.***

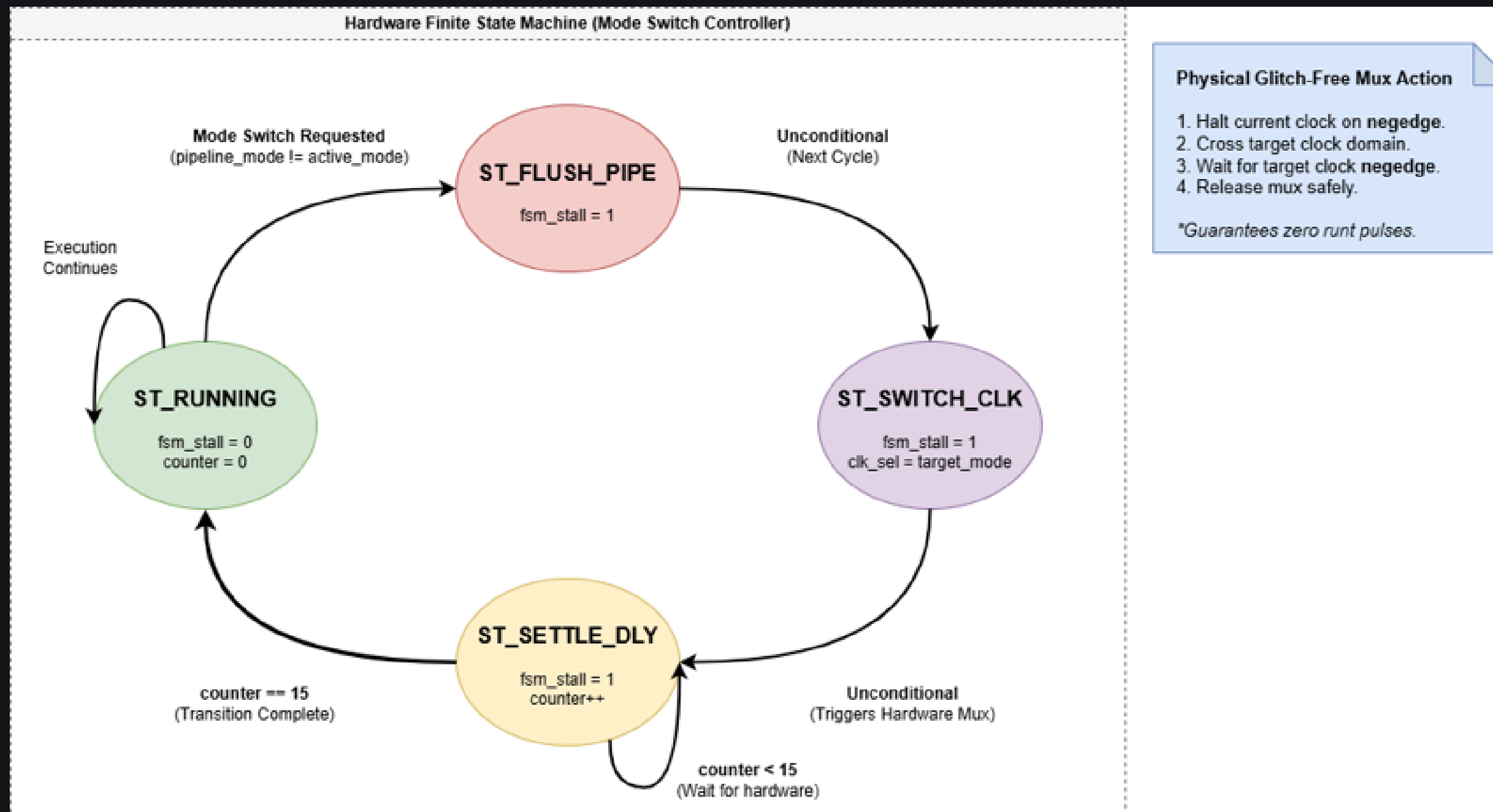
Operand Isolation (Combinational Power Gating)

While dual-clocking reduces frequency-based power, freezing operand propagation through the ALU during pipeline stalls or NOP bubbles eliminates useless gate toggling, ***lowering the switching activity factor to save dynamic energy.***



PHASE 3

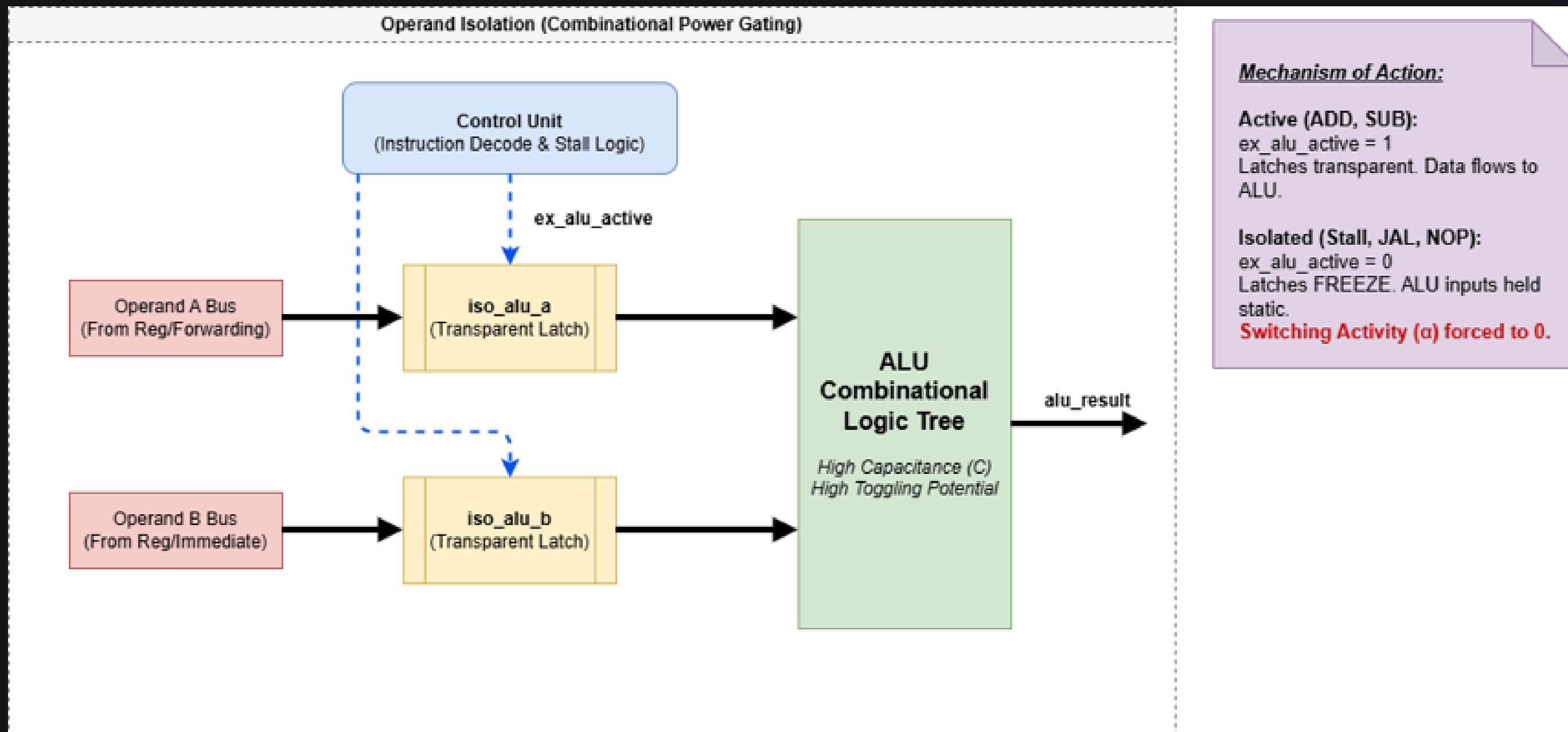
Glitch-Free Dual-Clock Domain Technique





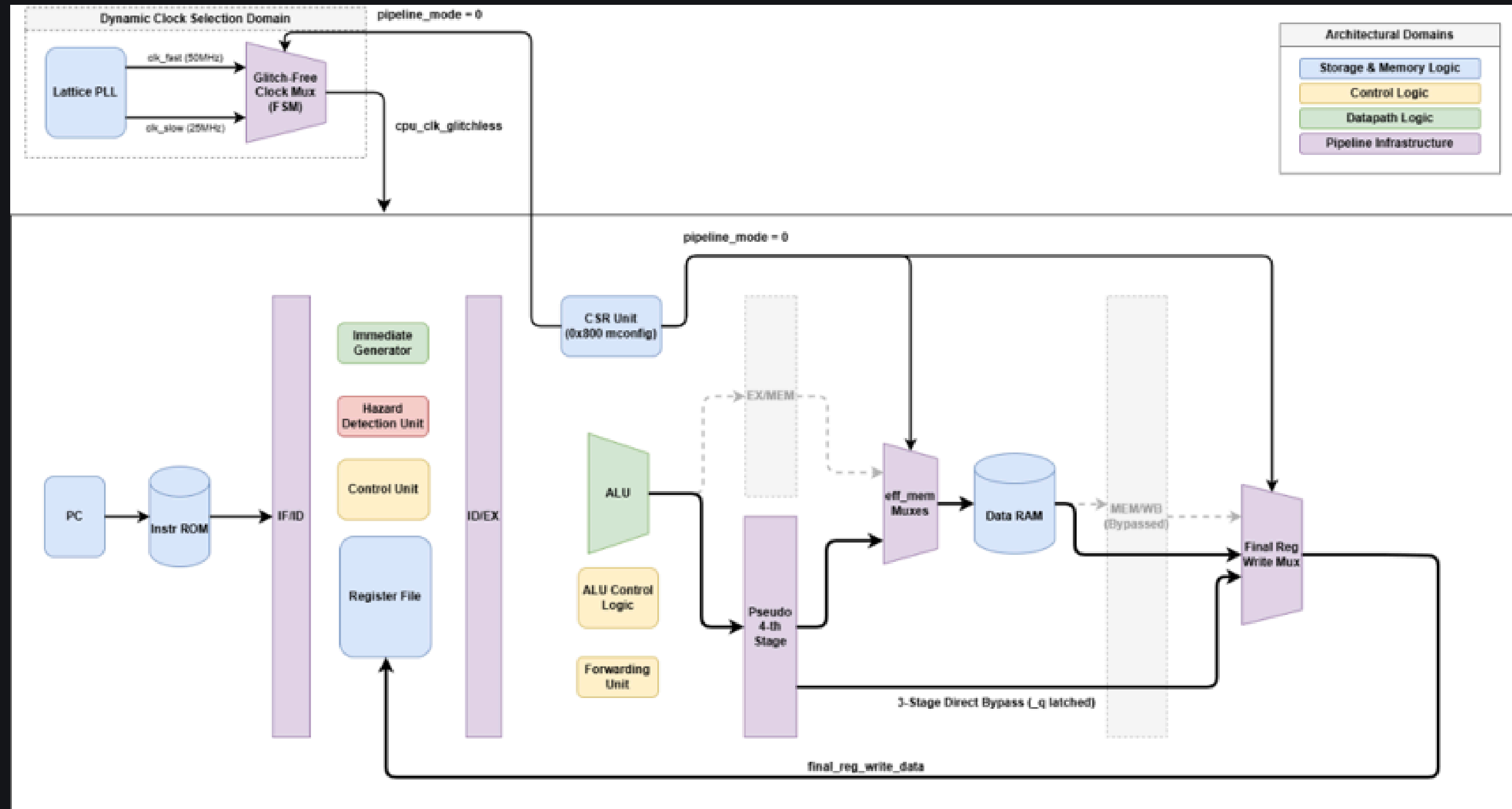
PHASE 3

Combinational Power Gating (Operand Isolation)

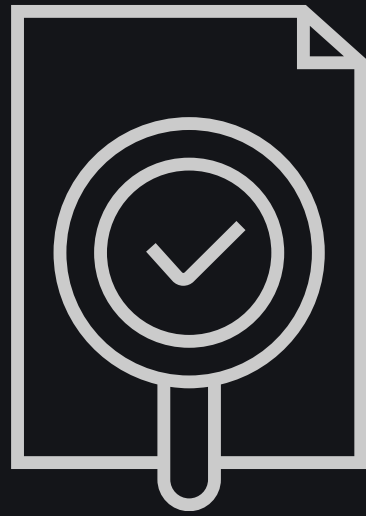




HOW IT WORKS WHEN IN 3-STAGE MODE



3-STAGE MODE

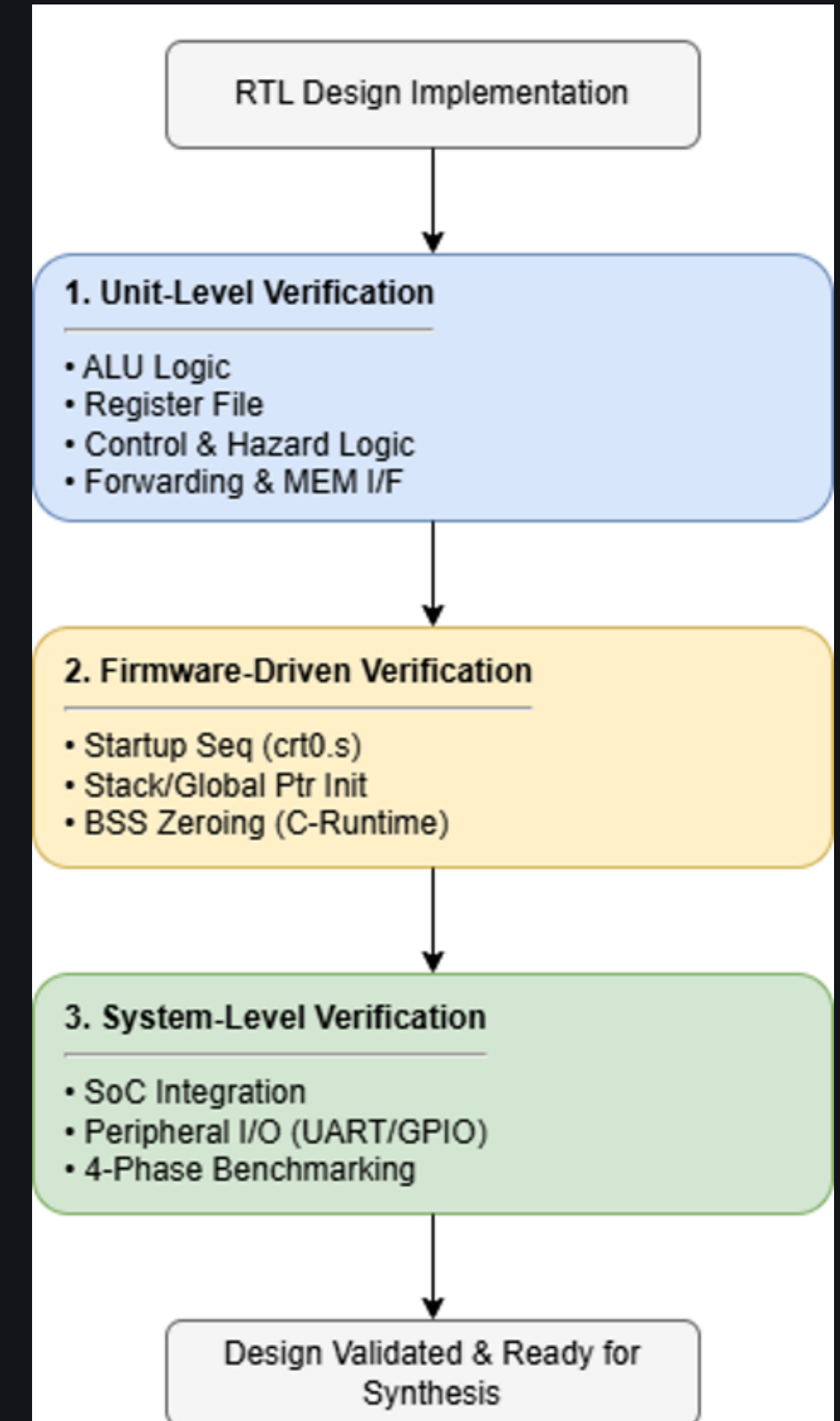


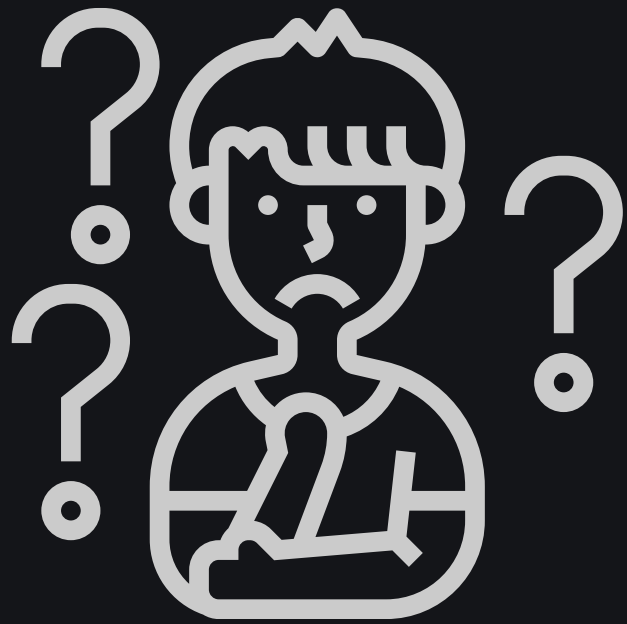
Verification

Phase 1

Phase 2

Phase 3





Problem Faced

1

Functional Correctness & Hidden Bugs

- Manual testing cannot easily uncover hidden architectural bugs, corner cases, or hidden data hazards introduced by the flexible, runtime-reconfigurable pipeline logic.

Conducted unit-level verification via isolated testbenches to validate the functional correctness of individual hardware modules (ALU, FSM, Hazard Unit).

2

Clock Glitches & Metastability

- Dynamically switching between two asynchronous clock domains (50 MHz and 20 MHz) using simple logic gates creates clock glitches and flip-flop metastability.

Developed a dedicated Mode Switch Controller driven by a 4-state Finite State Machine to safely intercept `pipeline\_mode` requests and smoothly orchestrate the clock domain crossings without hardware glitches.

RESULTS AND DISCUSSIONS



PHASE 1 - BASELINE SINGLE-CYCLE RV32I

RESOURCES USAGE

<i>Resource Types</i>	<i>Utilized Macro Cells</i>	<i>Total Available</i>	<i>Allocation Percentage (%)</i>
Look-Up Tables (LUT4)	2,348	84,255	2.78%
Dedicated Flip-Flops (DFF)	161	84,255	0.19%
Programmable Half-Adders	52	42,127	0.12%
Embedded Block RAM (EBR)	64	346	18.49%
Hardware Multipliers (DSP)	0	56	0%

POWER CONSUMPTION

<i>Power Parameter Component</i>	<i>Quantified Value (mW)</i>	<i>Budget Percentage (%)</i>
Static Baseline Leakage Power (P _{stat})	43.000 mW	71.67%
Active Dynamic Switching Power (P _{dyn})	16.000 mW	26.67%
Routing and Input/Output Pad (P _{io})	1.000 mW	1.66%
Total Core Power Dissipation (P _{total})	60.000 mW	100.00%

MAXIMUM OPERATING FREQUENCY: 29.109 MHz

PHASE 2 - FIXED 5-STAGE RV32I

RESOURCES USAGE

<i>Resource Types</i>	<i>Utilized Macro Cells</i>	<i>Total Available</i>	<i>Allocation Percentage (%)</i>
Look-Up Tables (LUT4)	2,899	84,255	3.44%
Dedicated Flip-Flops (DFF)	933	84,255	1.11%
Programmable Half-Adders	128	42,127	0.30%
Embedded Block RAM (EBR)	66	346	19.08%
Hardware Multipliers (DSP)	0	56	0%

POWER CONSUMPTION

<i>Power Parameter Component</i>	<i>Quantified Value (mW)</i>	<i>Budget Percentage (%)</i>
Static Baseline Leakage Power (P_{stat})	42.000 mW	50.00%
Active Dynamic Switching Power (P_{dyn})	41.000 mW	48.81%
Routing and Input/Output Pad (P_{io})	1.000 mW	1.19%
Total Core Power Dissipation (P_{total})	84.000 mW	100.00%

MAXIMUM OPERATING FREQUENCY: 54.347 MHz

PHASE 2 - FIXED 5-STAGE RV32I

Report generated on 2026-04-09 18:02 GMT by [riscov v](#)

Environment

Riscov Version	1.25.3
Riscv-arch-test Version/Commit Id	-
DUT	my core
Reference	spike
ISA	RV32IZicsr
User Spec Version	2.2
Privilege Spec Version	1.10

Yaml

[Show all details](#) / [Hide all details](#)

Name
/home/heng/fyp/work/my_core_isa_checked.yaml (show details)
/home/heng/fyp/work/my_core_platform_checked.yaml (show details)

Please visit [YAML specifications](#) for more information.

Summary

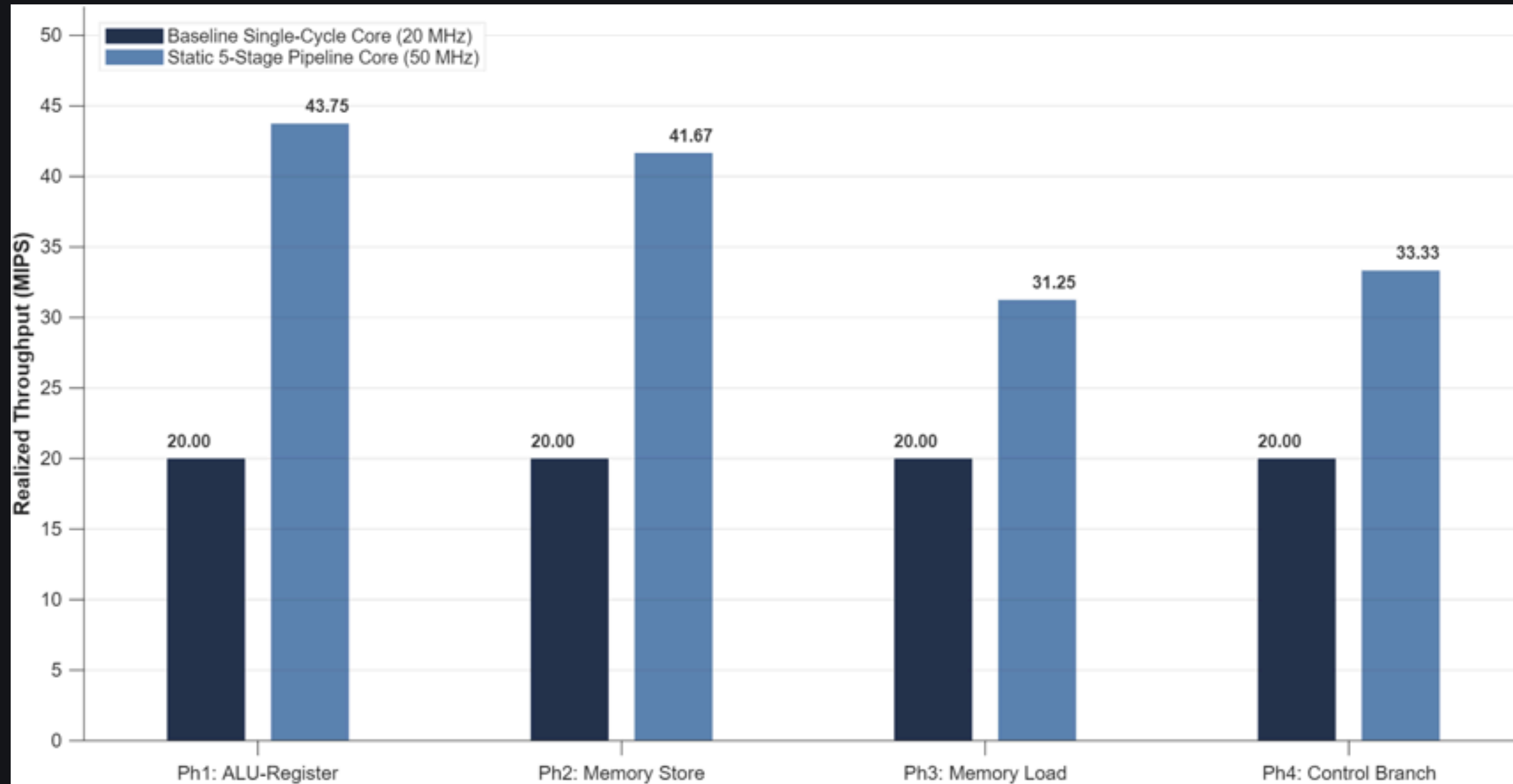
✓ 39Passed, ✓ 0Failed



WSL2: Windows Subsystem for Linux

To validate the pipelined 5-stage core was functionally correct before proceeding to adaptive core, formal verification was conducted using the RISCOF framework within WSL2. The 5-stage core achieved 100% compliance, passing all 39 standard RV32I architectural test suites by matching its memory signatures byte-for-byte against the Spike reference model.

PHASE 2 - FIXED 5-STAGE RV32I



Realized MIPS Throughput Comparison Between
Single-Cycle RV32I and Static 5-Stage RV32I

PHASE 3 - Software-Reconfigurable Adaptive RV32I

RESOURCES USAGE



<i>Resource Types</i>	<i>Utilized Macro Cells</i>	<i>Total Available</i>	<i>Allocation Percentage (%)</i>
Look-Up Tables (LUT4)	4,409	84,255	5.23%
Dedicated Flip-Flops (DFF)	2,011	84,255	2.39%
Programmable Half-Adders	184	42,127	0.44%
Embedded Block RAM (EBR)	66	346	19.08%
Hardware Multipliers (DSP)	0	56	0%

POWER CONSUMPTION



<i>Functional Block</i>	<i>5-Stage Mode (mW)</i>	<i>3-Stage Mode (mW)</i>	<i>Power Reduction (%)</i>
Gated State Registers	24.500 mW	0.000 mW	100.00%
Arithmetic Logic Unit	18.200 mW	6.100 mW	66.48%
Active Pipeline Control & Routing	16.300 mW	14.900 mW	8.59%
Total Active Dynamic Component (P _{dyn})	59.000 mW	21.000 mW	64.41%

MAXIMUM OPERATING FREQUENCY: 57.531 MHz (5-Stage)
24.914 MHz (3-Stage)

Cross-Architectural Efficiency Evaluation

$$E_{dyn} = \frac{P_{dyn}}{\text{Throughput (MIPS)}}$$

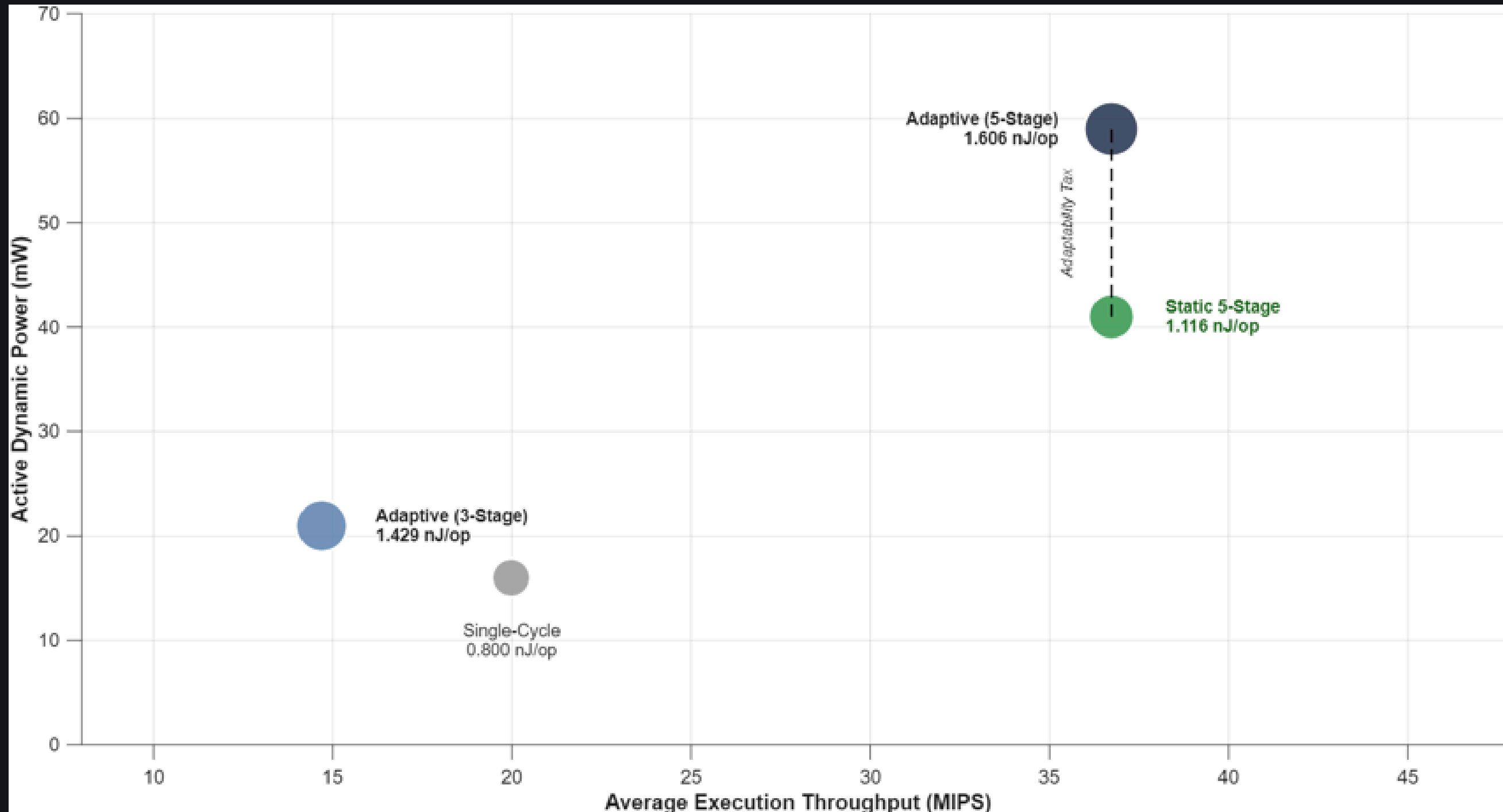
$$\text{Relative EDP Factor} \propto \frac{P_{dyn}}{\text{Throughput}^2}$$

Table 4.11: Cross-Architectural Figure of Merit Comparison

<i>Architecture</i>	<i>Single-Cycle</i>	<i>Static 5-Stage</i>	<i>Adaptive 5-Stage</i>	<i>Adaptive 3-Stage</i>
Operational Frequency	20.000 MHz	50.000 MHz	50.000 MHz	20.000 MHz
Dynamic Power (P _{dyn})	16.000 mW	41.000 mW	59.000 mW	21.000 mW
Throughput	20.000 MIPS	36.740 MIPS	36.740 MIPS	14.700 MIPS
Energy Per Instruction (E _{dyn})	0.800 nJ/instr	1.116 nJ/instr	1.606 nJ/instr	1.429 nJ/instr
Relative EDP Factor	0.040	0.030	0.044	0.097

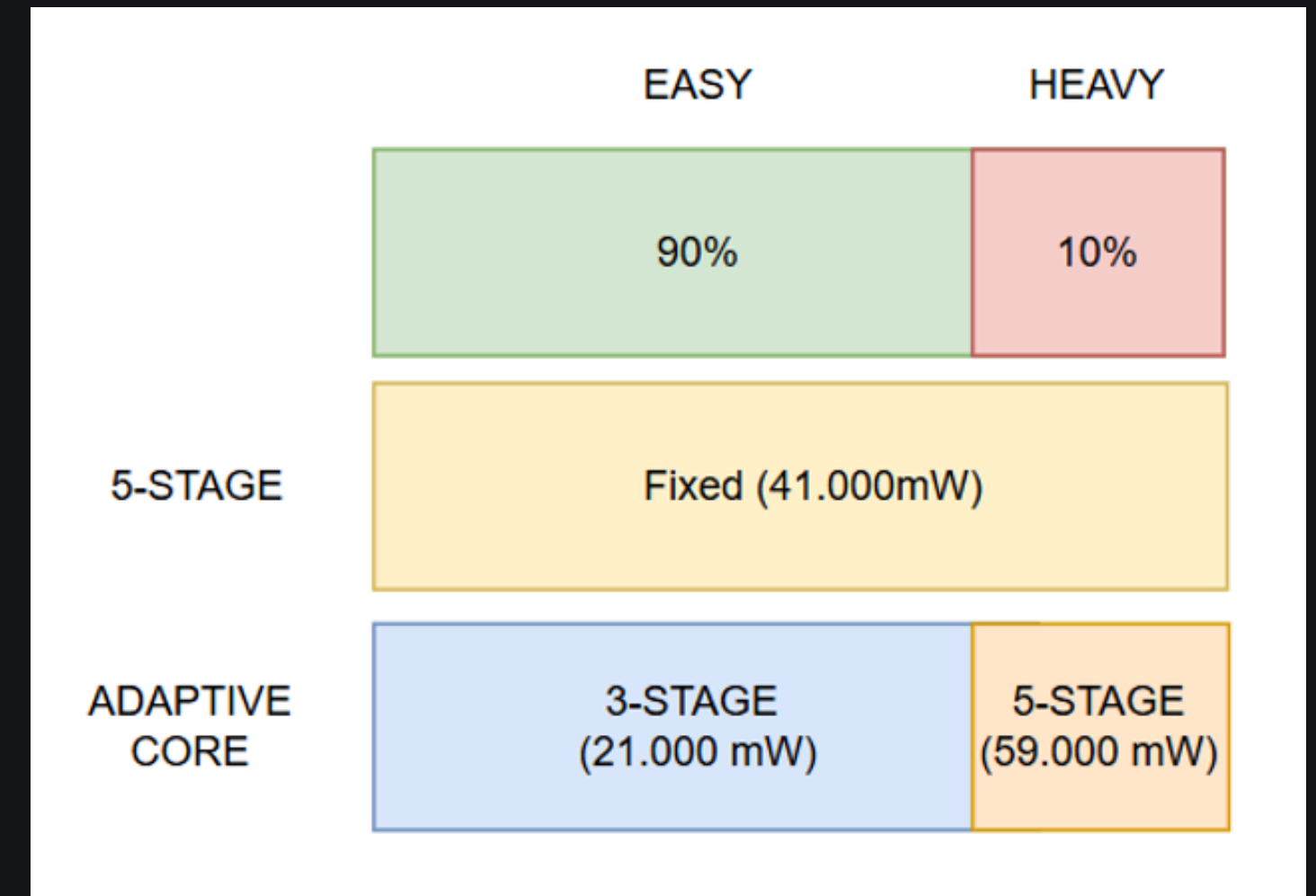
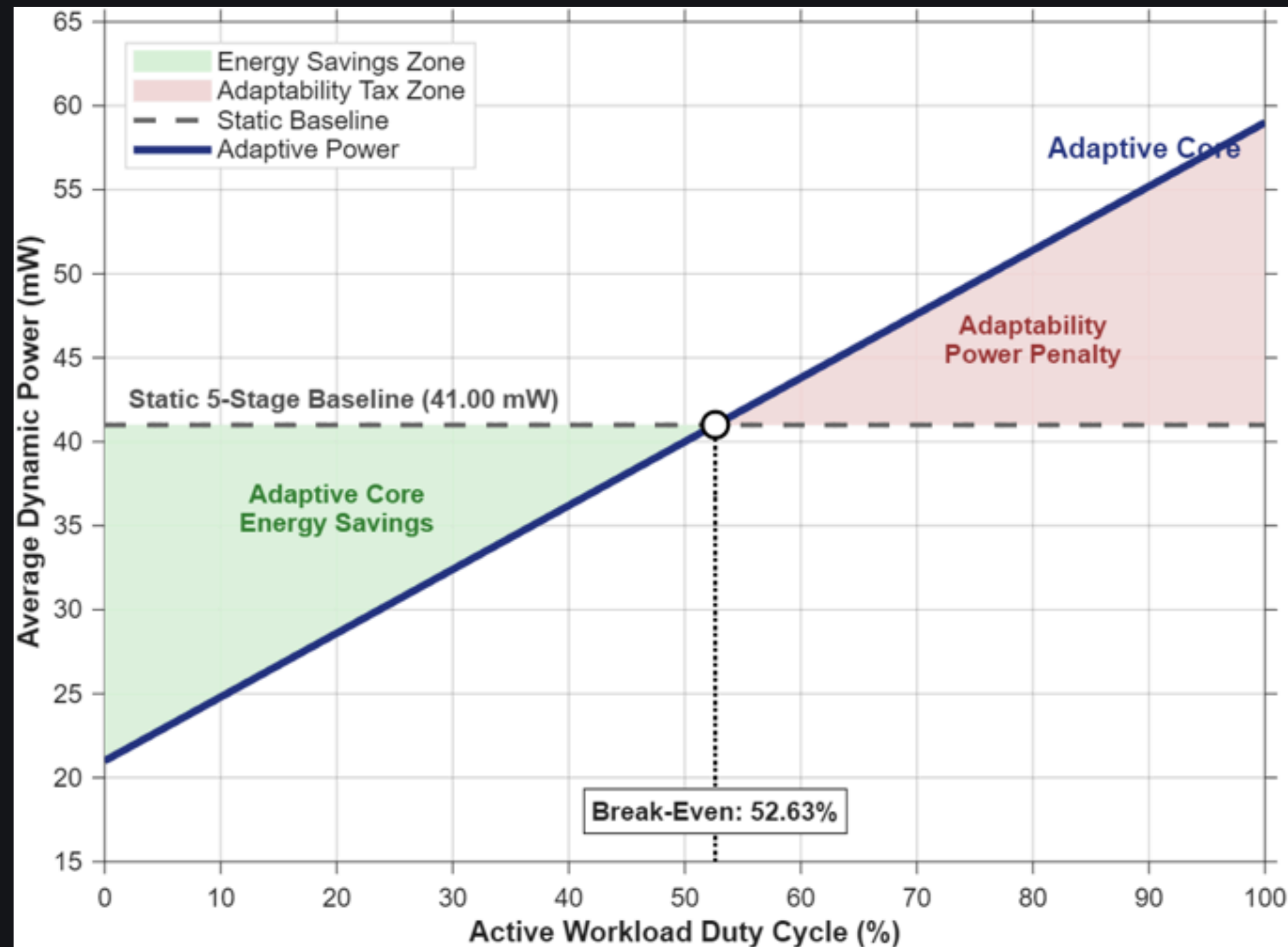
To account for the execution time penalty of a lower frequency, the architectures are evaluated using a relative, dimensionless Energy-Delay Product (EDP) factor where a lower value signifies a more optimal balance between power savings and performance loss.

Architectural Efficiency Comparison across Four Hardware Configurations

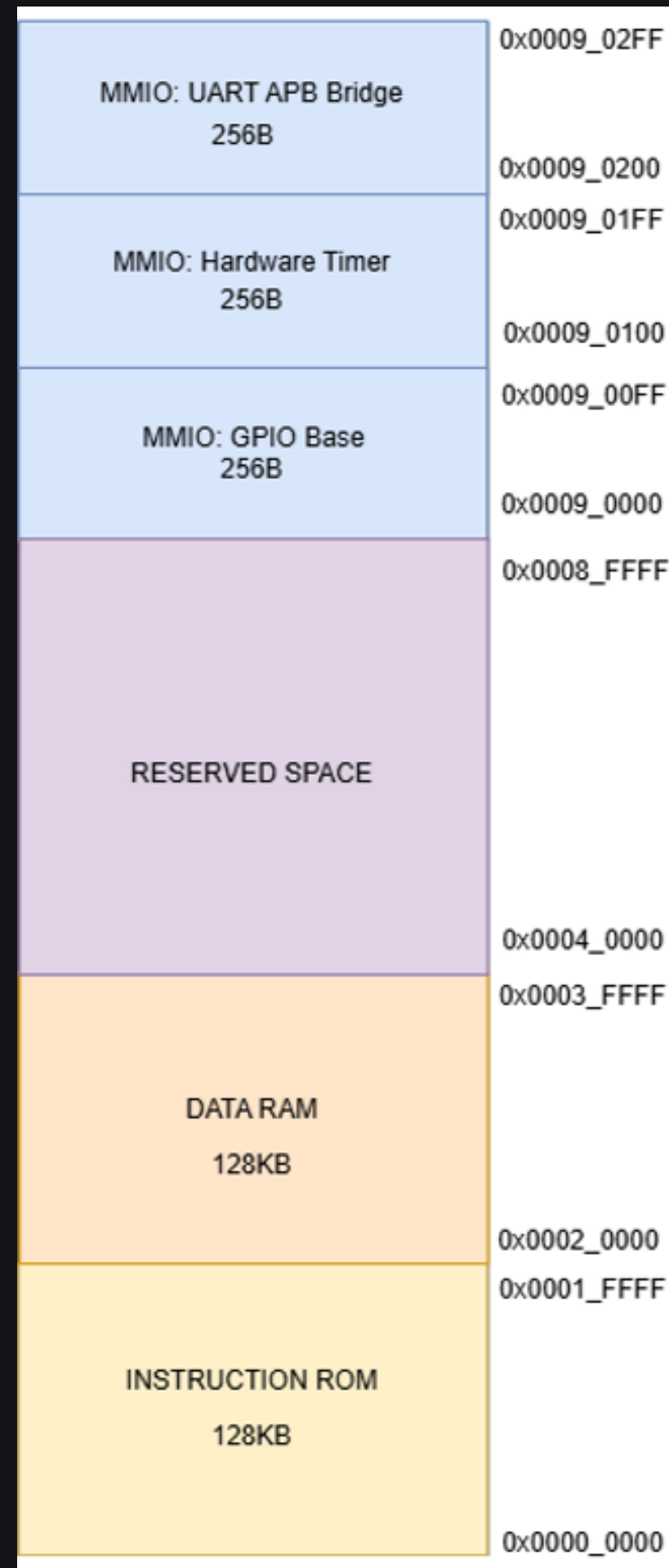


Average Dynamic Power vs Average Execution Throughput

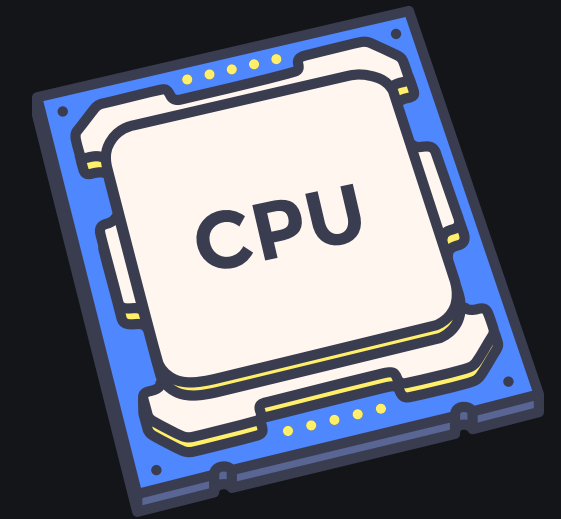
Relational Average Power Dissipation Models across Variable Active Duty Cycle



This workload boundary defines when each configuration is superior based on the active duty cycle.



Hardware Implementation



- **Hardcoded Memory Mapping:** A hardcoded memory-mapped architecture is utilized to manage data routing and prevent bus contention.
- **Defined Address Space:** The physical address space is explicitly divided between core memory structures and memory-mapped I/O (MMIO) peripheral zones.



DEMO

Future Works

- ***Integration of Standard Hardware Extensions:*** Expand the instruction set architecture beyond the base RV32I profile by integrating the standard RISC-V 'M' extension. This implements dedicated hardware multiplication and division units, eliminating the latency and execution overhead of software emulation libraries for compute-heavy tasks.
- ***Automated Hardware Performance Monitoring:*** Transition from manual, software-driven mode switches via explicit CSR writes to a fully automated hardware mechanism. Integrating an on-chip monitor to track real-time pipeline stall rates and instruction retirement metrics would allow the core to self-scale its execution depth dynamically.
- ***ASIC Migration and Fine-Grained DVFS:*** Migrate the RTL design from programmable FPGA fabric to a standard-cell ASIC flow to eliminate high static logic leakage. Moving to custom silicon enables true Dynamic Voltage and Frequency Scaling (DVFS), allowing the core voltage to scale down alongside the 20 MHz clock to maximize energy savings for ultra-low-power commercial IoT applications.

PROJECT SCHEDULE

[illegible]

CONCLUSION

- ***Adaptive RV32I Architecture:*** Successfully implemented a software-reconfigurable processor that dynamically transitions between a 5-stage performance mode and a 3-stage low-power mode to balance throughput and energy demands.
- ***Dual-Domain Clocking & Mitigation:*** Resolved the propagation delay of the collapsed 3-stage mode by using a custom 4-state FSM controller to safely drop the clock from 50 MHz to 20 MHz without glitches, backed by a hazard mitigation logic that achieved 100% RISCOF compliance.
- ***Switching Activity Optimization:*** Maximized dynamic energy savings by freezing operand propagation through the ALU during pipeline stalls and NOP bubbles, effectively eliminating useless gate toggling.
- ***Proven Efficiency Boundaries:*** Quantified a clear workload boundary at a 52.63% duty cycle. At a realistic 10% edge-computing duty cycle, the adaptive core cuts average power to 24.800 mW, yielding a 39.51% energy savings over the static baseline.

Q & A



THANK YOU